SOFTWARE REQUIREMENTS SPECIFICATION

ONLINE VOTING SYSTEM

Mini Project Proposal

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Technology Used

Technology	Use
Frontend	HTML, CSS, JavaScript
Backend	PHP
Database	MySQL
Local Server	XAMPP / Apache
Database Tool	phpMyAdmin
Editor	Visual Studio Code

1. Project Overview

The Online Voting System is a small web-based project for conducting an election through a website instead of using a manual voting process.

A voter can login, see the candidates and cast a vote. A candidate can apply to take part in the election, while the admin manages the election and checks the final result.

The main idea is to keep the complete election process in one place and make it simple to manage.

2. Why This System is Needed

In a manual election, voter details, candidate information, voting and vote counting are handled separately. This can take extra time and may create problems while maintaining records.

This project tries to solve these basic problems by keeping the information in a MySQL database and handling the voting process through PHP.

3. Users and Their Work

User	Main Work	Important Actions
Admin	Controls the election	Create election, manage candidates, approve applications, check voters and results
Voter	Takes part in voting	Register/login, view candidates, cast one vote and check result
Candidate	Contests the election	Apply, submit profile/manifesto, check approval and result

A candidate may also be an eligible voter if the rules of the particular election allow it. The exact rule can be decided while implementing the project.

4. How the System Will Work

1. The Admin creates an election and adds the required election details.
2. A person who wants to contest the election submits candidate details.
3. The Admin checks the candidate application and approves it.
4. Eligible voters register and login to the system.
5. Voters can see the active election and the approved candidates.
6. A voter selects one candidate and confirms the vote.
7. The system records the vote and prevents another vote in the same election.
8. After the election ends, the system counts the votes and shows the result.

ADMIN → CANDIDATE APPLICATION → APPROVAL → VOTING → VOTE COUNT → RESULT

5. Main Features

- Separate access for Admin, Voter and Candidate.
- User registration and login.
- Candidate application and approval.
- Election creation and management.
- Candidate details visible to voters.
- Online vote submission.
- One vote per voter for an election.
- Automatic vote counting.

- Final result display.

6. Basic Functional Requirements

Admin

- Login to the admin panel.
- Create, update and manage an election.
- Add, approve, reject or manage candidates.
- View registered voters.
- Check election status and results.

Voter

- Register and login.
- View active election and candidates.
- Cast a vote.
- Cannot submit another vote for the same election.
- View result when it is available.

Candidate

- Apply to contest an election.
- Submit candidate details and a short manifesto.
- Check approval status.
- View election information and final result.

7. Basic Quality Requirements

- Simple interface so that a new user can understand the voting steps.
- Passwords should be stored using hashing.
- Only authorized users should access the relevant dashboard.
- Voting records should be stored correctly in the database.
- The system should respond properly during registration, login and voting.
- The PHP files should be kept organized so the project can be updated later.

8. Proposed Database

MySQL will be used to store user, election, candidate and voting information. A simple proposed structure is given below.

Table	Main Fields	Purpose
users	id, name, email, password, role	Stores login and user information
candidates	id, user_id, position, manifesto, status	Stores candidate details and approval status
elections	id, name, description, start_date, end_date, status	Stores election details
votes	id, election_id, voter_id, candidate_id, voted_at	Stores submitted votes

10. Basic Security

- Password hashing.
- Login and session authentication.
- Role-based access to dashboards.
- Input validation.
- Duplicate-vote checking.
- Admin panel protection.

11. Expected Output

The final system should allow an election to be managed from one website. The expected flow is:

Admin creates election → candidates apply → admin approves candidates → voters login → voters select a candidate → vote is saved → election ends → votes are counted → result is displayed.

The project is expected to demonstrate practical use of PHP, MySQL, forms, sessions, database operations and role-based access.

12. Possible Future Improvements

- OTP or email verification.
- Two-factor authentication.
- Candidate profile photos.
- Charts for election results.
- Multiple election positions.
- Email notifications.
- Mobile application support.

13. Conclusion

The Online Voting System is planned as a simple and practical PHP-MySQL mini project. It will bring the main activities of an election into one system and reduce the amount of manual work involved in maintaining records and counting votes.

The project is small enough to implement within the mini-project scope while still covering useful concepts such as authentication, sessions, CRUD operations, database connectivity and voting logic.